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The two-dimensional cutting stock problem with usable leftovers and uncertainty in demand

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ABSTRACT

When dealing with cutting problems, the generation of usable leftovers proved to be a good strategy for decreasing material waste. Focusing on practical applications, the main challenge in the implementation of this strategy is planning the cutting process to produce leftovers with a high probability of future use without complete information about the demand for any ordered items. We addressed the two-dimensional cutting stock with usable leftovers and uncertainty in demand, a complex and relevant problem recurring in companies due to the unpredictable occurrence of customer orders. To deal with this problem, a two-stage formulation that approximates the uncertain demand by a finite set of possible scenarios was proposed. Also, we proposed a matheuristic to support decision-makers by providing good-quality solutions in reduced time. The results obtained from the computational experiments using instances from the literature allowed us to verify the matheuristic performance, demonstrating that it can be an efficient tool if applied to real-life situations.

1. Introduction

The Cutting Stock Problem (CSP) is a classical optimization problem widely studied in the scientific literature. The interest in this topic is due not only to its complexity, as it is an NP-hard combinatorial optimization problem, but also to its practical applicability in several industrial sectors, in which the minimization of raw material waste is fundamental for the economic performance of companies. In addition, cutting problems have interesting characteristics, such as their environmental impact (Coelho et al., 2017) and the dimensionality of the cut objects (Wäscher et al., 2007), which differentiate them from other operational research problems and allow the study of several variations with applications in practical situations.

We tackle the two-dimensional CSP, where the cut objects are rectangular plates. Also, leftovers can be generated for use in future cutting processes. This variation is called, in the literature, the Two-Dimensional Cutting Stock Problem with Usable Leftovers (2D-CSPUL). The problem consists of cutting standard plates (purchased from suppliers) to produce a demanded quantity of smaller rectangular items while optimizing an objective function (minimizing material waste or the number of cut plates), considering the possibility of generating usable leftovers during the cutting process. These leftovers are pieces whose dimensions meet pre-defined criteria that return to the stock and are not counted as waste. The criteria for defining the dimensions of these leftovers varies according to the specific characteristics of the agents involved, especially the history of orders received by the company. The generation of usable leftovers must be planned to increase the probability of their future use, minimizing the waste from a multiperiod perspective. This can provide several advantages for the company.

Although many studies related to usable leftovers have been developed, most of them are focused on the one-dimensional case, as seen in Cherri et al. (2014), which presents a review of the existing studies in the literature for the one-dimensional CSPUL (1D-CSPUL). Later, Arenales et al. (2015), Baykasoğlu and Özbel (2021), Khan et al. (2020), Liu et al. (2017), Ravelo et al. (2020), Tomat and Gradisar (2017) and Nascimento, de Araújo et al. (2022) also proposed solution methods for the 1D-CSPUL. Regarding the 2D-CSPUL, different approaches were presented by Andrade et al. (2016, 2014), Birgin et al. (2020), Clautiaux et al. (2019), Li et al. (2022) and Nascimento, Cherri et al. (2022).

Besides the dimension of plates, items and usable leftovers, the demand is also a relevant parameter that influences the quality of the solutions of the CSPUL. By allowing the generation of usable leftovers, the first question that arises is related to the use of these

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leftovers, which mainly depends on future demand that, in practical situations, may not be known when cutting plans are established. So many decisions are made, based on average values or approximating values through forecasting methods that use historical series, which do not provide accurate information, marring the planning for using leftovers.

The main contribution of this paper to the literature is to approach the 2D-CSPUL considering uncertainty in the demand for items and propose a new solution method with the expectation of better representing real situations. For two-dimensional problems, the present research is the first to investigate this topic. We used a two-stage approach to formulate the 2D-CSPUL under uncertainty. The two stages represent two planning periods: in the first, closer in time, demand is known; in the following stage, the demand is uncertain. First-stage decisions are taken knowing the demand for the first period but before uncertainty is defined for the second. These first-stage decisions impact the second stage since they include the generation of usable leftovers with the expectation of future use. In the second stage, the uncertain demand is approximated by a finite set of possible scenarios. Thus, the second-stage variables represent decisions to solve the problem associated with each scenario, adjusting the first-stage decisions and optimizing the use of leftovers.

Given the complexity of solving this exact integer problem, we propose a new matheuristic divided into three main steps: (1) create cutting patterns for all types of standard plates and leftovers in stock, (2) generate demand scenarios from a framework based on a co-evolutionary genetic algorithm, and (3) solve a stochastic mathematical model that determines the frequency of each cutting pattern in each scenario, minimizing the total waste. This stochastic model was solved by an exact solver and a non-exact strategy based on the L-shaped method, already recognized as efficient in solving stochastic linear problems. We evaluated the performance of the proposed matheuristic using computational tests with randomly generated instances from the literature. Our goal was to provide, in an acceptable computational time, satisfactory solutions that support the decision-making process in an uncertain context referring to the demand.

The remainder of this paper is structured as follows. Section 2 presents a literature review on the CSPUL and the CSP under uncertainties. In Section 3, we define the 2D-CSPUL with uncertainty in demand and present the proposed stochastic model to represent it. The developed matheuristic is described in Section 4, while Section 5 presents the results of the computational tests. Finally, conclusions are drawn and future studies are discussed in Section 6.

2. Literature review

Many studies have proposed different methods and strategies to solve the classical 2D-CSP. Regarding exact methods, Gilmore and Gomory (1965) and Kwon et al. (2019) used the column generation technique (Gilmore & Gomory, 1961, 1963). Christofides and Hadjiconstantinou (1995) developed a tree search algorithm using the State Space Relaxation (SSR) approach to reduce the number of nodes and improve the upper bounds of the problem. In Silva et al. (2010), the authors proposed an integer programming model and solved it through a commercial solver. Furini and Malaguti (2013) and Furini et al. (2016) used branch-and-price techniques to solve mixed-integer programming models. Martin et al. (2020) developed algorithms based on Benders decomposition and constraint-programming.

For non-exact methods, Wang (1983) and Vasko (1989) proposed combinatorial methods to build constrained cutting patterns from successive horizontal and vertical allocations of items. Lai and Chan (1997) developed an algorithm based on simulated annealing. Leung et al. (2001) also applied the simulated annealing heuristic, along with a genetic algorithm approach. In Cui (2005), the author presented dynamic programming algorithms to generate guillotine-cutting patterns with equal rectangles. Suliman (2006) proposed a three-step heuristic procedure that first creates minimum trim loss cutting patterns by solving one-dimensional problems and then determines their frequencies. Cintra et al. (2008), Furini et al. (2012), and Long et al. (2020) proposed methods based on dynamic programming and column generation techniques. Dusberger and Raidl (2015) also used dynamic programming to develop a heuristic, together with the Variable Neighborhood Search (VNS) method. In Bouaine et al. (2018), the authors used the Finite First Fit (FFF) and the Finite Best Fit (FBF) algorithms to develop a heuristic procedure. Wang et al. (2020) proposed a diving heuristic to obtain integer solutions from an integer programming formulation. Wu and Yang (2021) presented an iterative sequential heuristic procedure using recursive techniques. In Cheng et al. (2021), the authors developed a method based on the Symbiotic Organisms Search (SOS) algorithm, which is inspired by the interactive behavior among organisms in ecosystems.

Despite the extensive literature on the 2D-CSP, we tackled two variations that have never been studied simultaneously: the generation of usable leftovers and uncertainty in item demand. Below is the literature review of the main research on the CSPUL and the CSP under uncertainties. The approaches proposed in these studies helped us to face the challenges related to our problem.

2.1. Cutting stock problem with usable leftovers

The leftover generation strategy was first mentioned in Brown (1971), but not related to the CSP. For the 1D-CSPUL, Cherri et al. (2014) reviewed studies from the literature presenting applications of the problem, the mathematical formulations, comments on the results obtained from the computational experiments and proposals for future studies related to the 1D-CSPUL. Since then, Arenales et al. (2015) proposed a mathematical model to solve the 1D-CSPUL generating leftovers with pre-defined lengths in limited quantities. Coelho et al. (2017) presented a mathematical model and two heuristics to solve the 1D-CSPUL, assuming the generated leftovers can return to the stock or be sold to other companies. Liu et al. (2017) addressed the 1D-CSPUL considering the cut of conical bars for the production of tyres. A multi-objective mathematical model was proposed to minimize the total waste, the number of cutting patterns and the number of generated usable leftovers. Tomat and Gradisar (2017) proposed a solution method to find the optimal values for the stock size and the length of the leftovers with consecutive demands.

Still referring to the 1D-CSPUL, Ravelo et al. (2020) created a constructive heuristic to generate initial solutions for a meta-heuristic based on the GRASP strategy and an evolutionary algorithm to solve the problem. Khan et al. (2020) studied the 1D-CSPUL applied to the cutting of coils in a paper mill, considering a limited number of knives used in the cutting process. In Baykasoğlu and Özbel (2021), the authors presented a mixed integer linear programming model to solve the 1D-CSPUL applied to cutting marble blocks. In Nascimento, de Araújo et al. (2022), the authors studied the integration of the 1D-CSPUL with the Lot-Sizing problem considering that the demand for items is known and occurs within a finite planning horizon.

Regarding the 2D-CSPUL, Andrade et al. (2014) approached the non-guillotined 2D-CSPUL by presenting multilevel mathematical programming models. Andrade et al. (2016) presented two bilevel mathematical models to solve the non-exact guillotined 2D-CSPUL. The first model formulation is analogous to the bin-packing problem while the second model groups items of the same type. Clautiaux et al. (2019) addressed a variation of the 2D-CSPUL in which demand occurs consecutively and only the leftovers generated in the last cutting pattern for each order are used. Sumetthapiwat et al. (2020) also used column generation to solve a 2D-CSPUL formulation by considering that the dimensions of the usable leftovers were fixed and pre-defined.

Birgin et al. (2020) extended the formulation presented by Andrade et al. (2014) to consider the multiperiod problem. The proposed mathematical model generates up to two usable leftovers per plate from

vertical and horizontal guillotine cuts. Another paper that addressed the multiperiod 2D-CSPUL was Li et al. (2022), in which the authors solved the problem applied to a company that produces transformers by cutting steel coils. Nascimento, Cherri et al. (2022) proposed a mathematical model, which creates cutting patterns and determines their frequencies simultaneously, to represent the 2D-CSPUL and a heuristic procedure to solve large instances of the problem.

2.2. Cutting stock problems under uncertainty

Uncertain parameters in cutting processes motivated scientific research a few years after the first publications on cutting problems. As far as we know, Sculli (1981) was the first to consider uncertainties in cutting stock problems, presenting an exact method to solve the 1D-CSP with uncertainty in the dimension of the cut objects, using independently distributed normal random variables to represent defects at both ends of the object.

As with Sculli (1981), most published research has addressed the one-dimensional problem but with uncertainty in demand for the items. Beraldi et al. (2009) presented a stochastic formulation in two stages with recursion, in which possible scenarios for demand are considered. In the first stage of this formulation, the decision variables refer to the created cutting patterns. In the second stage, variables determine the frequencies of these patterns after the demand of each scenario is known.

Alem et al. (2010) used a two-stage nonlinear stochastic formulation to study 1D-CSP with random demand scenarios. The main difference from the approach of Beraldi et al. (2009) is the definition of decision variables at each stage. In Alem et al. (2010), the first stage variables are the frequencies of a set of pre-defined cutting patterns while the second stage variables represent corrective actions that adjust the decisions of the first stage, aiming to guarantee that the demand of each scenario is met. Wongprakornkul and Charnsethikul (2010) proposed a mathematical model for the 1D-CSP using discrete values to represent scenarios of uncertain demand. The authors considered a limited amount of cutting patterns to reduce the computational time for solving this strategy.

In Sarper and Jaksic (2018), the authors solved the 1D-CSP under uncertainty in demand using a multi-objective approach with priorities, where the main objectives are to minimize the underproduction of items and the number of overproduced items if there is overproduction. Sarper and Jaksic (2019) used a similar strategy, including simulations with the Monte Carlo method but with an initial focus on determining the optimal amount of objects in stock before knowing the demand for items.

Witthayapraphakorn et al. (2021) proposed an algorithm and integrated Benders decomposition and column generation methods to solve the 1D-CSP by considering that the demands occur in discrete, uncertain and time-varying values. Pitombeira-Neto and Murta (2022) formulated the 1D-CSP with uncertainty using the concept of Markov decision processes. This concept considers that the problem can be divided into states, and the transition from one state to another occurs according to decisions taken periodically. In the formulation of Pitombeira-Neto and Murta (2022), the states represent the number of items available in stock to meet an unknown demand. The decisions that lead to transitions refer to the number of objects cut according to each pre-defined cutting pattern.

Few studies have considered uncertainties in cutting problems with two or more dimensions. Krichagina et al. (1998) carried out a study applied to a paper mill where, in addition to defining a cutting plan to meet a random demand, a problem of scheduling the use of cutting machines is solved. The goal is to minimize the long-term costs of raw material waste, supply chain downtime, backorders, and item inventory. Alem and Morabito (2013) proposed three robust programming models to represent the integration of the 2D-CSP with the production planning problem considering uncertainty in demands and production costs.

Ide et al. (2015) also used robust optimization to address the cutting problem in a company that produced rectangular wooden components. The authors proposed a robust mathematical model which deals with uncertainties in the quality of the cut raw material. Both studies solved the proposed models through exact solvers. Fajemisin et al. (2023) addressed the integration of the CSP with the Vehicle Routing Problem (VRP) in the forest harvesting industry, which aims to exploit natural resources sustainably. A formulation for cutting problems of any dimension (one-, two-, or three-dimensional) was presented, considering uncertainties in the objects available in stock.

A stochastic approach for the one-dimensional cutting stock problem was recently studied by Cherri et al. (2023). In that paper, the authors proposed a mathematical model for the problem and used the column generation method to solve its linear relaxation. Computational experiments were performed using a multi-period framework to evaluate a deterministic and a two-stage stochastic approach solved on a rolling-horizon. The results showed that applying the stochastic approach decreased the total cost for all the tested instances. These results show the importance of explicitly considering uncertainty in cutting stock problems, even in simpler settings such as the one-dimensional problem.

In this work, we did not consider using the same solution method because two-dimensional problems are incomparably more complex to solve due to the geometric component that does not exist in the one-dimensional case. Therefore, a column generation approach would be inefficient since each sub-problem (pricing problem) requires solving a two-dimensional cutting problem with leftovers. In fact, column generation has been used in the literature for standard two-dimensional cutting stock problems, but only under very specific conditions that allow the use of efficient dynamic programming algorithms in the pricing problem.

To the best of our knowledge, the present paper is the first study tackling the 2D-CSPUL with uncertainty in the demand (2D-CSPULUD).

3. Problem definition and stochastic model

In the 2D-CSPULUD, a cutting plan must be defined to produce a set of ordered rectangular items by cutting rectangular larger plates available in stock. This planning process aims to optimize an objective function and must consider two operational aspects of the problem: *(i)* the possibility of generating usable leftovers that are kept in stock to be used in future cutting processes and *(ii)* the uncertainty in demand. As seen in the previous section, considering uncertain parameters in formulations of cutting problems is relevant and well accepted in the scientific literature. Despite the relevance, solution strategies that approximate uncertain parameters frequently use forecasting methods based on average values or the worst-case scenario. These strategies are easily implemented but provide limited contributions to the decision-making process, as they ignore an extensive set of possible scenarios for uncertainty.

The mathematical model for the 2D-CSPULUD was formulated as a two-stage stochastic problem. The first stage refers to the decisions made to meet a known deterministic demand, while the second stage determines the best cutting plan for a finite set of possible scenarios for the uncertain demand, adjusting the first stage decisions and optimizing the use of the generated usable leftovers. To illustrate the 2D-CSPULUD, consider an instance whose details are given in Table 1, with 2 types of standard plates in stock, each type with availability equal to 4 units, 3 types of items and 3 scenarios with the same probability of occurrence.

From a set of pre-defined cutting patterns, the decision variables x and y represent the frequencies of each cutting pattern in the first and second stages, respectively. Table 2 shows the detailed optimal solution for the instance. It presents the values for each stage and scenario, referring to the total area of cut plates (Column "Plates"),

Table 1
Data of the numerical example.

Plate	Width × Height	Stock
1	54 × 87	4
2	78 × 79	4
Item	**Width × Height**	**Demand (First stage)**
1	7 × 18	34
2	13 × 13	24
3	11 × 9	28
Scenario	**Probability of occurrence**	**Demand (Second stage)**
1	1/3	(22 28 34)
2	1/3	(28 40 21)
3	1/3	(22 36 28)

Table 2
Optimal solution of the numerical example.

Stage	Total area			
	Plates	Leftovers	Items	Waste
First	17 022	5460	11 112	**450 (2.64%)**
Second (Scenario 1)	16 320	5406	10 842	**72 (0.44%)**
Second (Scenario 2)	16 320	3432	12 367	**521 (3.19%)**
Second (Scenario 3)	15 054	3354	11 628	**72 (0.44%)**

generated usable leftovers (Column "Leftovers") and ordered items (Column "Items"). The last column shows the wasted area in the first stage and each scenario of the second stage. On all rows, the waste is obtained by subtracting the area of the generated usable leftovers and allocated items from the plate area. Therefore, the objective function value is equal to $450+72/3+521/3+72/3=671.67$. In this calculation, the waste in each scenario is multiplied by its probability of occurrence. Fig. 1 shows the cutting patterns and their respective frequencies.

Fig. 1 shows that the first stage solution generates two usable leftovers with equal dimensions (78 × 35), which are available to be used in each second stage scenario. The decisions in this scenario seek to adjust the first stage decisions, maximizing the use of the leftovers. As can be seen in the second stage, the leftovers were cut producing minimal waste.

The formulation of the 2D-CSPULUD also considers the control of the number of plates, items, and leftovers in stock. In the two-stage stochastic formulation, this limitation can encourage the cut of usable leftovers in the second stage since generating new leftovers depends on freeing up space in the stock. This situation happened in the solution of Fig. 1, especially in Scenario 1. As the first-stage solution generated two usable leftovers, the maximum amount allowed in stock, the second-stage solution could only produce leftovers if they cut those that already exist.

The two-stage stochastic mathematical model proposed to represent the 2D-CSPULUD, referred to as stochastic model M^{st}, receives as input parameters the sets J and J' of pre-defined cutting patterns and Ξ pre-determined second-stage demand scenarios. The set J is composed of the cutting patterns created for the plates available in stock at the beginning of the first stage and can be used in both stages. Set J' represents the cutting patterns built for the leftovers that can be generated in the first stage and are only available to be cut in the second stage.

In the stochastic model M^{st}, x_j are decision variables that indicate the frequency of each cutting pattern $j, j=1,\dots,J$, in the solution for the first stage. The decisions taken at this stage are common to all scenarios of the second stage, including those related to the usable leftovers generated. In the second stage, for each scenario $\xi, \xi=1,\dots,\Xi$, there are two sets of decision variables, $x'_{j\xi}$ and $y_{k\xi}$, which represent the frequencies of the cutting patterns j and $k, k=1,\dots,J'$, in the second stage. These variables adjust the decisions made in the first stage, meeting the demand of the scenarios and optimizing the use of the generated leftovers. To ensure the feasibility of the model, the demand constraints of both the first and second stages allow overproduction of items. However, the total area of surplus items is considered waste. Next, the complete stochastic model M^{st} is presented.

Indices

- $s=1,\dots,S$: number of types of standard plates in stock at the first stage.
- $r=1,\dots,R$: number of types of leftovers in stock at the first stage.
- $v=1,\dots,S+R$: number of types of plates (standard and leftovers) in stock at the first stage.
- $l=1,\dots,L$: number of types of leftovers that can be generated at the first stage.
- $i=1,\dots,I$: number of types of ordered items.
- $j=1,\dots,J$: number of cutting patterns for the plates in stock at the first stage.
- $k=1,\dots,J'$: number of cutting patterns for the leftovers that can be generated at the first stage.
- $\xi=1,\dots,\Xi$: number of scenarios.

Parameters

- W_v: width of a plate of type v.
- H_v: height of a plate of type v.
- e_v: number of plates of type v in stock at the first stage.
- W_l^L: width of a leftover of type l.
- H_l^L: height of a leftover of type l.
- w_i: width of item i.
- h_i: height of item i.
- d_i: demand for item i in the first stage.
- $d'_{i\xi}$: demand for item i in scenario ξ in the second stage.
- a_{ij}: number of items type i in cutting pattern j.
- a'_{ik}: number of items type i in cutting pattern k.
- $b_{jv}=\begin{cases}1, & \text{if the cutting pattern } j \text{ cuts a plate of type } v\\ 0, & \text{otherwise.}\end{cases}$
- $b^L_{kl}=\begin{cases}1, & \text{if the cutting pattern } k \text{ cuts a leftover of type } l\\ 0, & \text{otherwise.}\end{cases}$
- $g_{jl}=\begin{cases}1, & \text{if the cutting pattern } j \text{ generates a leftover of type } l\\ 0, & \text{otherwise.}\end{cases}$
- U: maximum number of leftovers in stock.
- α: reuse rate of generated leftovers.
- β_ξ: probability of occurrence of scenario ξ.
- p_j: waste from cutting a plate according to pattern j.
  - $p_j=\sum_{v=1}^{S+R} b_{jv}W_vH_v-\alpha\sum_{l=1}^{L} g_{jl}W_l^LH_l^L-\sum_{i=1}^{I} a_{ij}w_ih_i.$
- q_k: waste from cutting a leftover according to pattern k.
  - $q_k=\alpha\sum_{l=1}^{L} b^L_{kl}W_l^LH_l^L-\sum_{i=1}^{I} a'_{ik}w_ih_i.$

Variables

- x_j: number of plates cut according to pattern j at the first stage.
- $y_{j\xi}$: number of plates cut according to pattern j in scenario ξ at the second stage.
- $y'_{k\xi}$: number of leftovers cut according to pattern k in scenario ξ at the second stage.

Stochastic model M^{st}

Minimize:

$$\sum_{j=1}^{J} p_jx_j+\sum_{i=1}^{I}\Big(\sum_{j=1}^{J} a_{ij}w_ih_ix_j-d_iw_ih_i\Big)+$$
$$\sum_{\xi=1}^{\Xi}\beta_\xi\Big[\sum_{j=1}^{J} p_jy_{j\xi}+\sum_{k=1}^{J'} q_ky'_{k\xi}+\sum_{i=1}^{I}\Big(\sum_{j=1}^{J} a_{ij}w_ih_iy_{j\xi}$$

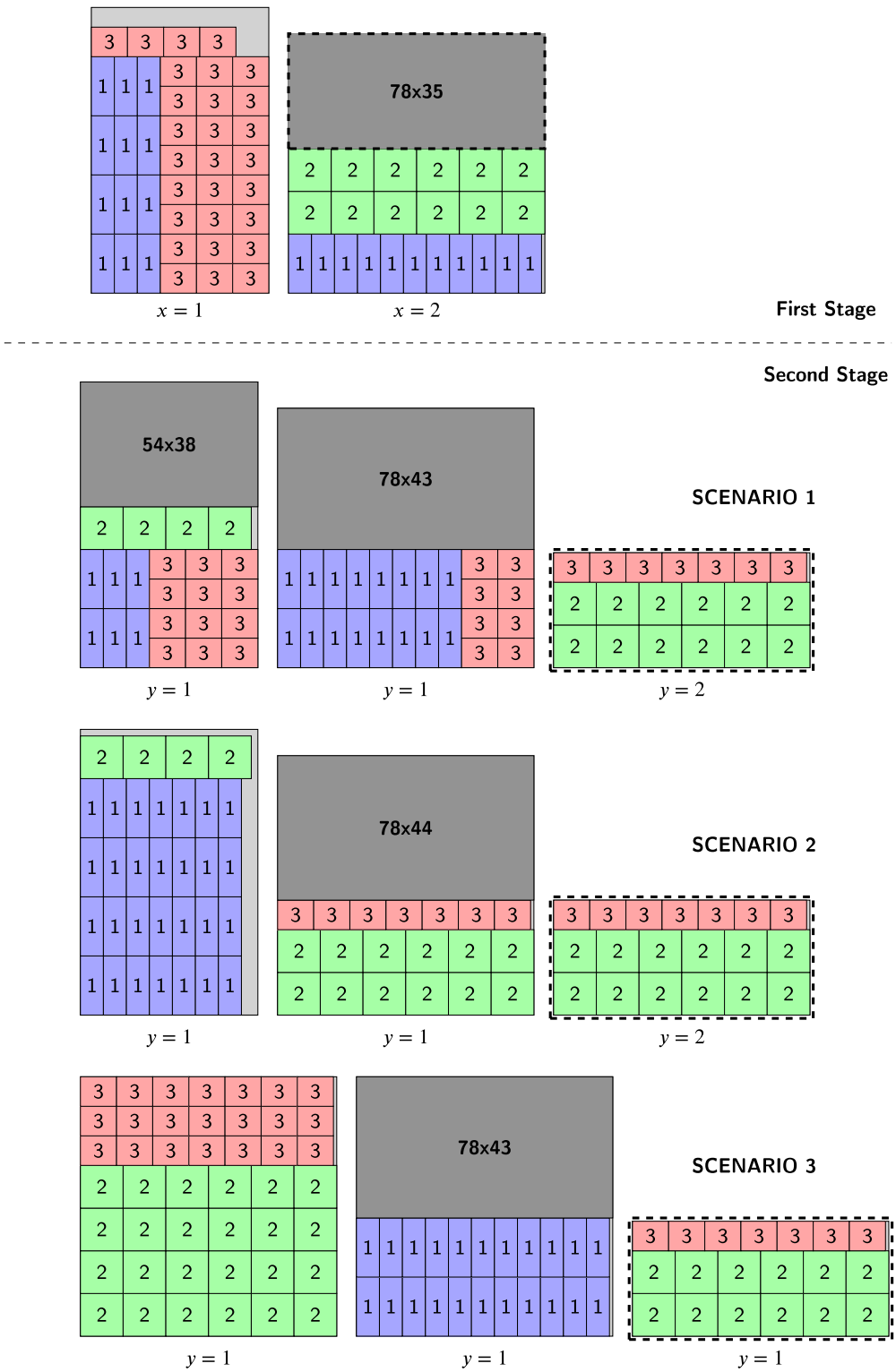

Fig. 1. Cutting patterns used in the optimal solution of the numerical example.

$$+\sum_{k=1}^{J'} a'_{ik} w_i h_i y'_{k\xi} - d'_{i\xi} w_i h_i)] \tag{1}$$

Subject to:

$$\sum_{j=1}^{J} a_{ij} x_j \geq d_i, \quad \forall i \tag{2}$$

$$\sum_{j=1}^{J} b_{jv} x_j \leq e_v, \quad \forall v \tag{3}$$

$$\sum_{j=1}^{J} \sum_{l=1}^{L} g_{jl} x_j - \sum_{j=1}^{J} \sum_{r=1}^{R} b_{j,S+r} x_j \leq U - \sum_{r=1}^{R} e_{S+r} \tag{4}$$

$$\sum_{j=1}^{J} a_{ij} y_{j\xi} + \sum_{k=1}^{J'} a'_{ik} y'_{k\xi} \geq d'_{i\xi}, \quad \forall i, \xi \tag{5}$$

$$\sum_{j=1}^{J} b_{jv} x_j + \sum_{j=1}^{J} b_{jv} y_{j\xi} \leq e_v, \quad \forall v, \xi \tag{6}$$

$$\sum_{k=1}^{J'} b^L_{kl} y'_{k\xi} - \sum_{j=1}^{J} g_{jl} x_j \leq 0, \quad \forall l, \xi \tag{7}$$

$$\sum_{j=1}^{J} \sum_{l=1}^{L} g_{jl} x_j - \sum_{j=1}^{J} \sum_{r=1}^{R} b_{j,S+r} x_j + \sum_{j=1}^{J} \sum_{l=1}^{L} g_{jl} y_{j\xi} - \sum_{j=1}^{J} \sum_{r=1}^{R} b_{j,S+r} y_{j\xi} - \sum_{k=1}^{J'} y'_{k\xi} \leq U - \sum_{r=1}^{R} e_{S+r}, \quad \forall \xi \tag{8}$$

$$x_j \in Z^+, \quad \forall j, \tag{9}$$

$$y_{j\xi} \in Z^+, \quad \forall j, \xi, \tag{10}$$

$$y'_{k\xi} \in Z^+, \quad \forall k, \xi. \tag{11}$$

In the stochastic model M^{st}, the objective function (1) minimizes the total waste obtained in the production of the ordered items in the first and second stages. The sums $\sum_{i=1}^{I}(\sum_{j=1}^{J} a_{ij} w_i h_i x_j - d_i w_i h_i)$ and $\sum_{i=1}^{I}(\sum_{j=1}^{J} a_{ij} w_i h_i x'_{j\ xi} + \sum_{k=1}^{J'} a'_{ik} w_i h_i y'_{k\xi} - d'_{i\xi} w_i h_i)$ ensure that the area of the items produced in excess is counted as waste. Constraints (2) and (5) refer to the fulfillment of demand in the first and second stage, respectively. Constraints (3) and (6) limit the stock of the plates available at the beginning of the first stage, while constraint (7) refers to the stock of usable leftovers generated in the first stage and cut in the second stage. The limit for usable leftovers in stock is given by (4) and (8). Domain of the variables are presented in (9)–(11).

Despite the interest in finding exact solutions, due to the complexity in solving the 2D-CSPULUD for real-sized instances that generally have a large number of constraints and integer decision variables referring to the building of cutting patterns and their frequencies, a matheuristic is also proposed to solve the problem.

4. Matheuristic

The proposed matheuristic was developed to find solutions as close to the optimal values as possible while maintaining an acceptable computational time. It is divided into three steps. In Step 1, a set of cutting patterns is created for all types of plates in stock. Then, a scenario generation method for the demand of items uses these cutting patterns in Step 2. This method is based on an evolutionary framework proposed by Oliveira et al. (2019) that uses the BRKGA - Biased Random-Key Genetic Algorithms strategy, which proved to be adequate for the problem under consideration since it evolves parallel scenarios without needing complete information about uncertainty. Finally, in Step 3, the stochastic model M^{st} is solved by an exact solver and through a newly proposed heuristic adapted from the L-shaped method. Fig. 2 illustrates the matheuristic structure, and pseudocode Algorithm 1 describes the procedures called during its execution. These procedures are presented in the following subsections.

Algorithm 1: Matheuristic main program.

```
Input: Boolean exact^sol, heuristic^sol;
cutting^patterns = Algorithm 2 ();
demand^scenarios = Co-evolutionary Algorithm (cutting^patterns);
if exact^sol == TRUE then
    Solve stochastic model M^st (cutting^patterns, demand^scenarios);
end
if heuristic^sol == TRUE then
    Algorithm 3 (cutting^patterns, demand^scenarios);
end
```

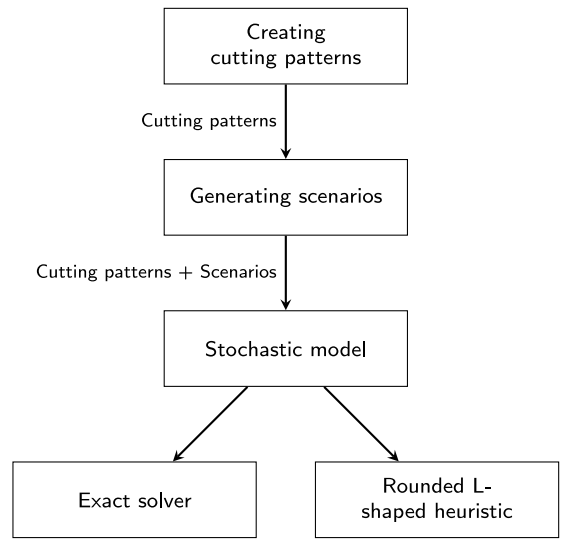

Fig. 2. Proposed matheuristic for the 2D-CSPUL under uncertainty.

4.1. Creating cutting patterns

To create cutting patterns with and without usable leftovers for the 2D-CSPUL under uncertainty, two mathematical models, M^{CP} and M^L, were proposed and solved iteratively. These models are based on the strategy proposed by Nascimento, Cherri et al. (2022), which only considers guillotine cuts. Guillotine cuts divide the rectangular panels into two parts as they are applied orthogonally from one edge to the other. Guillotine cuts are a very common constraint in the industry due to both cutting machine technology and production cycle time considerations, as patterns with guillotine cuts are quicker to perform due to their lower complexity compared to patterns with non-guillotine cuts.

The strategy proposed by Nascimento, Cherri et al. (2022) also combines items to generate new types of items, called macro-items. Macro-items aim to minimize material waste by enhancing the diversity of items, making it possible to create better cutting patterns and find better solutions. For a set of ordered items, sorted by descending order of their heights ($h_1 \geq h_2 \geq h_3 \geq \cdots \geq h_I$), K types of macro-items are defined by allocating two or more items one above the other. Parameter n_{ik} indicates the number of items i in the macro-item $k, k = 1, \ldots, K$. The height of a macro-item type k (h^K_k) is the sum of the heights of all items on it, i.e., $h^K_k = \sum_{i=1}^{I} n_{ik} h_i$. The maximum height allowed for a macro-item is the height of the highest ordered item. The width of a macro-item type k (w^K_k) is equal to the largest width among all items on it, i.e., $w^K_k = \max\{w_i \mid n_{ik} > 0\}$.

The cutting patterns are built by dividing the cut plate into horizontal strips. The height of each strip is defined from a set of H^F possible values $h^F_\eta, \eta = 1, \ldots, H^F$, including the different heights of items and macro-items. Considering F_s the maximum number of strips in cutting patterns for a plate type s, if a height η is assigned to a strip $f, f = 1, \ldots, F_s$, i.e., the strip is initialized, then items and macro-items can be allocated to f on the right side of the last item already allocated since their heights are lower or equal than h^F_η.

To maximize the diversity of cutting patterns and, consequently, the quality of the solutions, cutting patterns are created for all possible combinations of ordered items, i.e., all the different subsets containing at least one type of item from the original set. For example, if a problem instance has I = 3 types of ordered items, there will be seven possible subsets: {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3} and {1, 2, 3}. Therefore, the number of possible combinations is defined as $C = 2^I - 1$. When creating a new cutting pattern for a plate s according to combination

$c, c = 1, \dots, C$, the models M^{CP} and M^L will receive the parameters related only to the types of items in c.

The model M^{CP}, which creates cutting patterns without usable leftovers, is defined next.

Indices

- $i = 1, \dots, I'$: number of types of ordered items in the combination;
- $k = 1, \dots, K'$: number of types of macro-items with items in the combination;
- $f = 1, \dots, F$: number of strips for the cut plate;
- $\eta = 1, \dots, H^F$: number of heights for strips.

Parameters

- W: width of the cut plate;
- H: height of the cut plate;
- w_i: width of item i;
- h_i: height of item i;
- A_i: maximum number of items i allocated to a strip;
- A_k^N: maximum number of macro-items k allocated to a strip;
- w_k^K: width of macro-item k;
- h_k^K: height of macro-item k;
- n_{ik}: number of items i in macro-item k;
- h_η^F: height η for strips;
- $m_{i\eta}$: parameter defined as 1 if the height of item i is equal to or lower than height η; otherwise, it is defined as 0;
- $m_{k\eta}^K$: parameter defined as 1 if the height of macro-item k is equal to or lower than height η; otherwise, it is defined as 0.

Variables

- $z_{\eta f} = \begin{cases} 1, & \text{if height } \eta \text{ is assigned to strip } f \\ 0, & \text{otherwise.} \end{cases}$
- a_{if}: number of items i allocated to strip f;
- a_{kf}^K: number of macro-items k allocated to strip f;

Model M^{CP}

Maximize:

$$\sum_{i=1}^{I'} \sum_{f=1}^{F} h_i w_i a_{if} + \sum_{i=1}^{I'} \sum_{k=1}^{K'} \sum_{f=1}^{F} n_{ik} h_i w_i a_{kf}^K \tag{12}$$

Subject to:

$$\sum_{\eta=1}^{H^F} z_{\eta f} \le 1, \quad \forall f \tag{13}$$

$$\sum_{\eta=1}^{H^F} \sum_{f=1}^{F} h_\eta^F z_{\eta f} \le H \tag{14}$$

$$\sum_{i=1}^{I'} w_i a_{if} + \sum_{k=1}^{K'} w_k^K a_{kf}^K \le W, \quad \forall f \tag{15}$$

$$a_{if} \le A_{iv} \sum_{\eta=1}^{H^F} m_{i\eta} z_{\eta f}, \quad \forall i, f \tag{16}$$

$$a_{kf}^K \le A_{kv}^N \sum_{\eta=1}^{H^F} m_{k\eta}^K z_{\eta f}, \quad \forall k, f \tag{17}$$

$$\sum_{f=1}^{F} a_{if} + \sum_{k=1}^{K'} \sum_{f=1}^{F} n_{ik} a_{kf}^K \ge 1, \quad \forall i \tag{18}$$

$$z_{if} \in \{0, 1\}, \quad \forall i, f \tag{19}$$

$$a_{if} \in [0, A_i] \text{ and integer}, \quad \forall i, f \tag{20}$$

$$a_{kf}^K \in [0, A_k^N] \text{ and integer}, \quad \forall k, f \tag{21}$$

In the M^{CP} model, the objective function (12) maximizes the allocation of items and macro-items into the cutting pattern. Constraint (13) guarantees that only one height is assigned to each initialized strip. Constraints (14) and (15) refer to the height and width of the cut plate, respectively. Constraints (16) and (17) prevent initialized strips receiving items and macro-items with heights greater than their heights. Constraint (18) ensures that at least one item of each type existing in the combination is allocated to the cutting pattern. The domains of the variables are presented in (19)–(21).

The second proposed model M^L creates cutting patterns that produce usable leftovers. Following the same strategy used by Nascimento, Cherri et al. (2022), each cutting pattern generates only one usable leftover through a horizontal guillotine cut, i.e., the leftover has the same width as the standard plate and the height varies within a pre-defined interval $[h_{min}^R H, h_{max}^R H]$, where h_{max}^R and h_{min}^R are values between 0 and 1, and $h_{max}^R > h_{min}^R$. The decision variable q represents the height of the usable leftover.

In the M^L model, the objective function maximizes the total area of items and macro-items allocated to the cutting pattern plus the area of the generated usable leftover. The formulation of the objective function is given by:

$$\text{Max}: \quad \sum_{i=1}^{I} \sum_{f=1}^{F} h_i w_i a_{if} + \sum_{i=1}^{I} \sum_{k=1}^{K} \sum_{f=1}^{F} n_{ik} h_i w_i a_{kf}^K + W q \tag{22}$$

Constraints (13) and (15)–(21) from the M^{CP} model are also present in the M^L model. However, constraint (14) is replaced by (23), which subtracts the height of the usable leftover from the height of the cut plate:

$$\sum_{\eta=1}^{H^F} \sum_{f=1}^{F} h_\eta^F z_{\eta f} \le H - q \tag{23}$$

Consider the stock with S types of standard plates and R types of leftovers. As only standard plates can generate usable leftovers, the M^{CP} model is solved for each pair (v, c), $v = 1, \dots, S + R$, while the M^L model is solved for each (s, c), $s = 1, \dots, S$. Parameter J indicates the number of cutting patterns created for the set of plates available in stock at the beginning of the first stage. The maximum value for J is defined as $J_{max} = (2 \times S \times C) + (R \times C)$. It is not possible to guarantee that $J = J_{max}$ for all 2D-CSPUL problems. Since it depends on the dimensions of items and plates, it may be unfeasible to solve the M^{CP} and M^L models for some combinations of items.

From the set J, it is possible to list the L types of leftovers that can be generated in the first stage. For each leftover $l, l = 1, \dots, L$, the M^{CP} model is solved iteratively, disregarding constraint (18), for N times, creating the best cutting pattern different from those already built in previous iterations. Parameter J' represents the total number of cutting patterns created for leftovers, and it is calculated as $J' = N \times L$. Pseudocode Algorithm 2 describes this procedure.

To illustrate the proposed strategy, consider the instance described in Table 1, which has only standard plates in stock ($S = 2$) and $I = 3$ types of items. For this instance, there are $C = 7$ item combinations: *(i)* {1}, *(ii)* {2}, *(iii)* {3}, *(iv)* {1, 2}, *(v)* {1, 3}, *(vi)* {2, 3}, and *(vii)* {1, 2, 3}. Therefore, fourteen cutting patterns are created for each plate in stock (seven with and seven without usable leftovers), totaling $J = 28$ patterns. Figs. 3 and 4 show the cutting patterns created for Plate 1 of the given instance.

The cutting patterns in Fig. 4 allow the enumeration of the different types of usable leftovers which can be generated in the first stage. Although seven cutting patterns generate usable leftovers from Plate 1, three produce the same leftover, with dimension 54 × 51. Therefore, there are $L = 5$ different types of usable leftovers. Fig. 5 shows the $J' = 5$ cutting patterns that would be created for these leftovers if $N = 1$. The cutting patterns created by this strategy are used in the second step of the matheuristic, described in the next section.

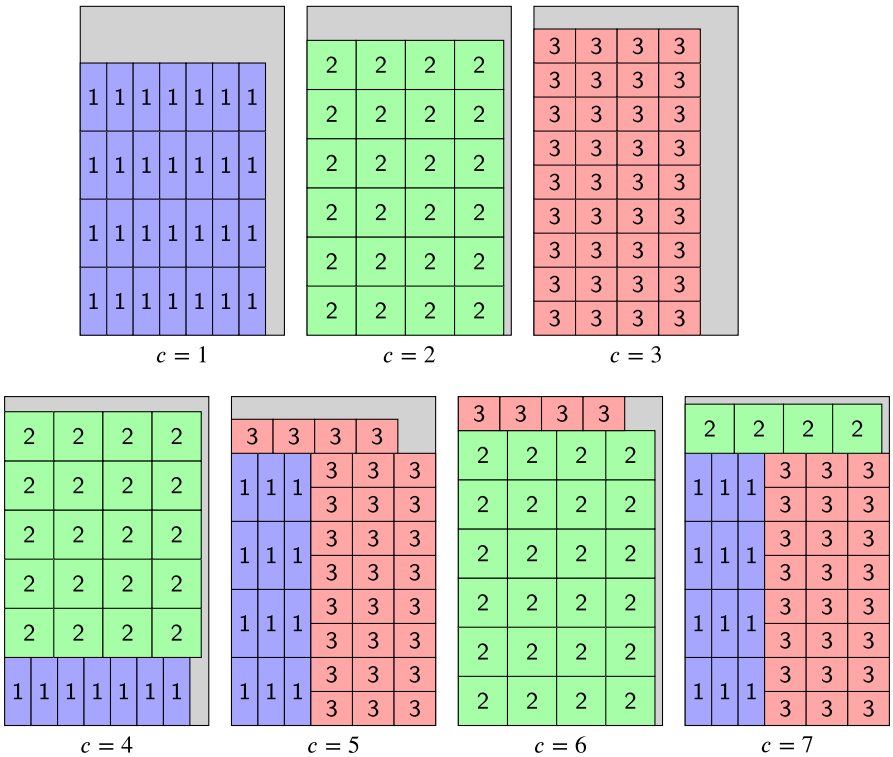

Fig. 3. Best cutting patterns, without usable leftovers, for every combination of items.

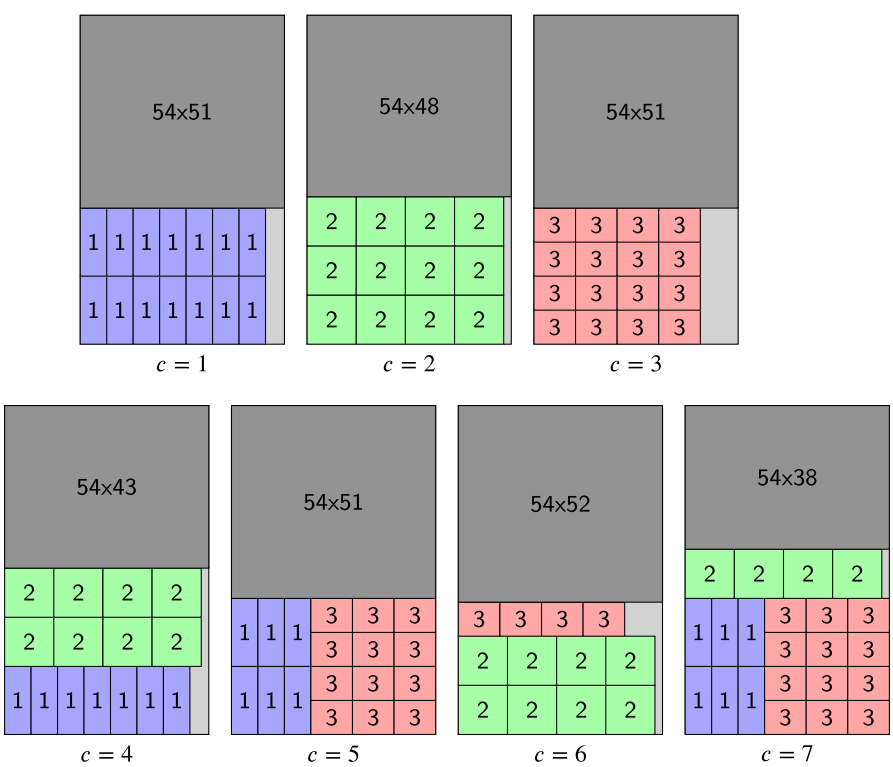

Fig. 4. Best cutting patterns, with usable leftovers, for every combination of items.

4.2. Generating scenarios for demand of items

In the second step of the proposed matheuristic, a set of scenarios for item demand is defined using a co-evolutionary algorithm proposed by Oliveira et al. (2019). This algorithm evolves, simultaneously, a population of scenarios and a population of solutions. Each individual in the solution population represents a solution for the first stage. Each individual in the scenario population represents a realization of the item demand in the second stage. Both populations are related through a *fitness evaluation*. The fitness values of each solution measures the quality of its performance against the scenario population, i.e., what is the overall value of the solution when faced with each scenario. The

Algorithm 2: Strategy for creating cutting patterns.

```
J = J' = L = 0;
for c = 1...C do
    for s = 1...S do
        Solve model M^CP for (s, c);
        if Feasible problem then
            J = J + 1;
        end
        Solve model M^L for (s, c);
        if Feasible problem then
            J = J + 1;
            if A new type of usable leftover is generated then
                L = L + 1;
            end
        end
    end
    for r = 1...R do
        Solve model M^CP for (r, c);
        if Feasible problem then
            J = J + 1;
        end
    end
end
for l = 1...L do
    for n = 1...N do
        Solve model M^CP for l disregarding constraint (18);
        if New cutting pattern is created then
            J' = J' + 1;
        end
    end
end
```

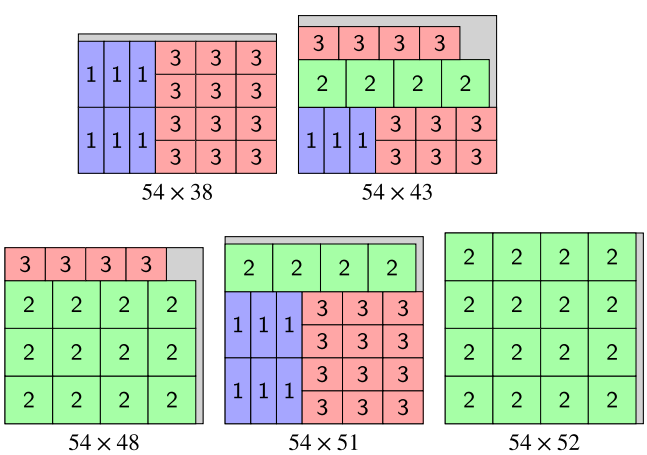

Fig. 5. Best cutting patterns for each usable leftover that can be generated in the first stage.

scenarios fitness values quantify their contribution to the population diversity according to their impact on the performance of the solution. The evolutionary strategy of this co-evolutionary algorithm is based on a Biased Random-Key Genetic Algorithm - BRKGA (Gonçalves & Resende, 2011, 2013), which belongs to the class of Genetic Algorithms.

In BRKGA, a given number of solutions to an optimization problem are represented by a population of *chromosomes*, which are vectors of real numbers between 0 and 1. Each number in these vectors is called a *gene*. By means of a *decoder*, or *decoding procedure*, the chromosomes are converted into solutions to the problem, while a *fitness function* evaluates the quality of these solutions. Starting from an initial population of randomly generated chromosomes, the objective of BRKGA is to iteratively evolve the population of solutions, with each new generation of solutions being composed of three categories of chromosomes:

- *Elite*: chromosomes from the current population whose solutions present the best values in the fitness evaluation.
- *Mutant*: new randomly generated chromosomes.
- *Offspring*: new chromosomes generated from the *crossover* process, which consists of combining the genes of elite and non-elite chromosomes from the current population. In this process, the decision about from which parent the chromosome offspring will inherit each gene is biased, as the probability of inheriting from the elite chromosome is always greater than 50%.

After defining a new population, the algorithm returns to the chromosome decoding step and follows the same procedure to evolve the next population. This procedure is repeated until some stopping criterion is met. These criteria may refer to the number of evolved generations or the quality of current solutions.

The co-evolutionary framework proposed by Oliveira et al. (2019) expands the strategy of the BRKGA method by evolving two populations of chromosomes simultaneously. This co-evolutionary approach is fit for solving optimization problems with uncertainty since it considers a second population to represent scenarios for the uncertain parameter. In applying the co-evolutionary algorithm to the two-stage formulation for the 2D-CSPULUD, each solution chromosome represents a solution for the first stage, while each scenario chromosome represents a possible occurrence for the demand of the items in the second stage. The two populations are related by $FN(solution, scenario)$, which indicates the value of each first-stage solution in each scenario. The $FN(solution, scenario)$ values are used in the fitness evaluation to measure the quality of the solution chromosomes. Also, they quantify the contribution of the scenario chromosomes to the population diversity according to their impact on the performance of a given solution.

An *API* (Application Programming Interface) developed by Oliveira et al. (2022) is used to implement the co-evolutionary algorithm and evolve scenarios for the 2D-CSPULUD. This application requires only the programming of the problem-dependent stages of the co-evolutionary BRKGA, i.e., the decoders applied to the chromosome populations and the calculation of the $FN(solution, scenario)$ values.

4.2.1. Solution decoding procedure

The proposed approach for the solution chromosomes decoder determines the first-stage decisions from a set of cutting patterns created by the strategy described in Section 4.1. Considering that, for a given instance of the 2D-CSPULUD, J cutting patterns were created for the plates available in stock at the beginning of the first stage, each solution chromosome is composed of $2 \times J$ genes. The first J genes, $g_1, \ldots, g_J$, are sorted in ascending order and indicate the sequence of use of the cutting patterns. And the last J genes, $g'_1, \ldots, g'_J$, determine their frequencies.

Following the order established by the first J genes, the first cutting pattern $j, j = 1, \ldots, J$ is selected and an upper bound UB_j for its frequency is defined. Parameter UB_j is equal to the minimum among the following values, ensuring the solution remains feasible:

- Number of plates cut by pattern j available in stock.
- Minimum number of plates that need to be cut according to j to meet the demand of at least one of the ordered items.
- Maximum number of new leftovers allowed in stock (if j generates a usable leftover).

After defining UB_j, the interval between 0 and 1 is divided into $UB_j + 1$ equal parts, each part representing a possible value for the frequency x_j. The value assigned to x_j is defined according to the subinterval in which the gene g'_j is inserted. Then, the stock and demand are updated and cutting pattern $j+1$ is selected. This process is repeated until all cutting patterns have been analyzed or the demand for all items is met.

4.2.2. Scenario decoding procedure

Each chromosome in the scenario population represents a possible scenario for the demand in the second stage. Assuming I is the number of types of ordered items, each scenario chromosome is composed of I genes $(d'_1, \dots, d'_I)$. By setting the parameters d'_{min} and d'_{max} the minimum and maximum values for the demand of items, the interval between 0 and 1 is divided into $d'_{max} - d'_{min} + 1$ equal sub-intervals, each sub-interval associated with a possible demand value. The demand for each item i is defined according to the sub-interval in which the gene d'_i is inserted.

4.2.3. Fitness evaluation

After decoding all chromosomes from both populations, the quality of each solution chromosome is verified in front of each scenario chromosome. Each solution-scenario pair contains the decisions for the first stage and the decisions for the second stage, associated with the scenario chromosomes, are defined through a heuristic procedure.

For each solution chromosome so, a value $Fitness^1_{so}$ is calculated. This value represents the total area used by the solution associated with so and is calculated as $Fitness^1_{so} = A^S - \alpha \times A^L + P$. In this expression A^S is the total area of cut plates, A^L is the total area of generated leftovers, and P are the penalties applied if the solution does not meet the demand for at least one of the ordered items. The parameter $0 \leq \alpha \leq 1$ indicates the reuse rate of leftovers, which limits their impact on the objective function since it is not possible to guarantee they will be entirely used in the second stage.

The value $Fitness^1_{so}$ is an individual evaluation metric for the solution chromosomes. To evaluate the performance of each chromosome so against each scenario ce, the second stage sub-problem is solved and must meet the demand of ce. The stock of this problem is composed of the plates not used in the first stage and the usable leftovers generated by so. The proposed heuristic procedure solves the second stage of each pair (so, ce) using the J and J' cutting patterns created by the strategy described in Section 4.1. The steps of this heuristic are detailed below.

- ***Step (1):*** Sort the $J + J'$ cutting patterns by descending order of the number of different item types produced. If two or more cutting patterns produce the same number of item types, priority will be given to those that generate the lowest waste percentage.
- ***Step (2):*** Following the order established in Step 1, select the first cutting pattern j that has not been analyzed and produces only items whose demands have not been met.
- ***Step (3):*** Set the frequency of j equal to the minimum value among *(i)* the number of plates available in stock of the type cut by j; *(ii)* the maximum number of leftovers allowed in stock (if j produces a usable leftover), and *(iii)* the minimum number of times j must be used to meet the demand for at least one of the item types.
- ***Step (4):*** Update the stock and demand. If the demand for all items has been met or there are no remaining cutting patterns to be analyzed, stop the procedure. Otherwise, return to Step 2.

The value $Fitness^2_{so,ce}$ represents the total area used by the solutions obtained for the second stage referring to each pair (so, ce). They are calculated as $Fitness^2_{so,ce} = A^S + \alpha \times (A^G - A^L) + P$. The only difference compared to calculating the $Fitness^1_{so}$ values is the inclusion of the area of usable leftovers generated by the first-stage solution and cut in the second stage (A^G).

From $Fitness^1_{so}$ and $Fitness^2_{so,ce}$, the $FN(so, ce)$ values are calculated and represent the total waste percentage of the solutions for each pair (so,ce) considering the two stages of the problem. Assuming that A^I is the total area of ordered items in both the first and second stage, $FN(so, ce)$ is calculated as:

$$FN(so, ce) = \frac{100 * (Fitness^1_{so} + Fitness^2_{so,ce} - A^I)}{(Fitness^1_{so} + Fitness^2_{so,ce})}$$

After being calculated, the $FN(so, ce)$ values are used by the API in the fitness evaluation process. The fitness of each scenario chromosome measures its contribution to the population diversity, considering the impact on the performance of the solution chromosomes. Regarding the fitness of each solution chromosome so, it is defined by the Laplace criterion, which considers the non-weighted average value of $FN(so, ce)$ in all scenarios.

4.3. Solution methods for the M^{st} stochastic model

From the cutting patterns and the demand scenarios generated, to solve the M^{st} stochastic model, an exact and a non-exact approach are used. Initially, it is solved through an exact solver satisfying all the integrality constraints of the decision variables. However, solving it under these conditions can be impracticable due to the exponential number of decision variables, especially for real-sized instances. In this context, a heuristic procedure is proposed. This procedure consists of *(i)* using the *L-shaped* method to obtain a solution with relaxed second-stage variables and *(ii)* solving the second stage separately for each scenario, fixing the first-stage variables in the values of the solution obtained by the L-shaped method.

Since it was proposed by Van Slyke and Wets (1969) for solving two-stage stochastic problems, the L-shaped method has been widely used in the literature, showing the ability to obtain optimal solutions with reduced computational effort compared to the direct solution of the monolithic model. The L-shaped method decomposes the two-stage stochastic problem into two parts. Firstly, it solves a *primary problem* to find a set of first-stage decisions considering an outer linearization of the second stage. The primary problem can be defined as follows:

$$\text{Min: } z = c^T x + \theta \tag{24}$$

Subject to:

$$Ax = b$$

$$D_f x \geq d_f, \quad f = 1, \dots, F \tag{25}$$

$$E_o x + \theta \geq e_o, \quad o = 1, \dots, O \tag{26}$$

$$x \geq 0$$

$$\theta \in R \tag{27}$$

In the objective function (24), θ represents an approximation of the second-stage value function. Constraints (25) and (26) are respectively known as *feasibility cuts* and *optimality cuts*. These cuts are added to the primary problem during the execution of the L-shaped algorithm. A feasibility cut is generated when the first-stage solution obtained by the primary problem is infeasible in the second-stage sub-problem. Once the feasibility of the first-stage solution is verified, the linear sub-problem related to the second-stage decisions is solved for each scenario ξ to check its optimality. Although the L-shaped method is efficient and appropriate for the problem being addressed, its use to solve the M^{st} stochastic model is impracticable since it is impossible to obtain the dual variables from the sub-problems considering the integrality constraints of the second-stage decision variables. Thus, we proposed the *Rounded L-shaped* heuristic, described in the next section.

4.3.1. Rounded L-shaped heuristic

The Rounded L-shaped heuristic has two steps. Assuming M^{st}_{relax} is a version of the M^{st} stochastic model in which the integrality constraints of the second stage decision variables are relaxed, M^{st}_{relax} is solved through the L-shaped method in the first step. In the second step, assuming M^{st}_{fix} is a version of the M^{st} stochastic model in which the first-stage decision variables are fixed to the values of the solution obtained in the first step, M^{st}_{fix} is solved separately for each scenario using an exact solver. At the end of the procedure, the second-stage variables referring to all scenarios will be integer-valued. The pseudocode "Algorithm 3" describes the Rounded L-shaped heuristic.

Algorithm 3: *Rounded L-shaped Heuristic*

Solve M^{st}_{relax} through the L-shaped method;
for $\xi = 1...\Xi$ **do**
  Solve M^{st}_{fix} only for scenario ξ;
end

5. Computational tests

This section presents the description and the results of the computational experiments conducted to evaluate the performance of the proposed mathematical model and matheuristic to solve the 2D-CSPULUD. All the heuristics and models used in the different stages of the proposed matheuristic were implemented in the Optimization Programming Language (OPL) language and solved by the IBM ILOG CPLEX, version 20.10 The API used in the scenario evolution stage, coded in C++, was executed in Visual Studio 2022. The tests were carried out on a computer with an Intel Core i7 processor, 2.60 GHz, 8 GB of RAM, and Windows 10 operating system.

Tests were performed with 16 classes of randomly generated instances taken from Nascimento, Cherri et al. (2022) and each class is composed of 10 instances. The classes of instances established $S = 2$ types of standard plates and $R = 1$ or 2 types of leftovers initially in stock. The number of types of ordered items varied as $I = 3$, 5, 8 or 10. Assuming H^{max} and W^{max} are, respectively, the maximum height and width among the standard plates in stock, two categories were considered for the dimensions of items:

- *Small* (Z), whose heights and widths were generated in the intervals $[0.05 \times H^{max}, 0.25 \times H^{max}]$ and $[0.05 \times W^{max}, 0.25 \times W^{max}]$.
- *Medium* (M), whose heights and widths were generated in the intervals $[0.10 \times H^{max}, 0.50 \times H^{max}]$ and $[0.10 \times W^{max}, 0.50 \times W^{max}]$.

The demands for items, here considered as the demands in the 2D-CSPULUD first-stage problem, were defined in the interval [5, 50]. The name of each class follows the format "I/Dimension of items/S/R". For example, "5/Z/2/2" refers to a class with 5 types of small items, 2 types of standard plates, and 2 types of leftovers in stock.

The cutting patterns for all instances were created using the strategy described in Section 4.1. The minimum and maximum heights for generated leftovers were defined as $h^R_{min} = 0.4$ and $h^R_{max} = 0.6$. The value $N = 5$ was defined for each type of usable leftover that can be generated by the first-stage solution.

Next, the co-evolutionary BRKGA algorithm was executed considering the number of chromosomes in the solution and scenario populations equal to 100 and 50, respectively. Both populations, composed of 70% descendent chromosomes, 20% elite, and 10% mutants, evolved over 1000 generations. For the crossover process, we set the probability of inheriting each gene from the elite chromosome equal to 70%. These are the default parameters for the API proposed in Oliveira et al. (2022) for the co-evolutionary BRKGA algorithm. This configuration has been validated since the BRKGA was proposed (Gonçalves & Resende, 2011, 2013) and has shown consistent performance across different problems and applications.

Table 3 presents the results referring to the first two steps of the proposed matheuristic. This table shows, for each class of instances, the average numbers of cutting patterns created for the plates initially in stock (Column J), types of usable leftovers that can be generated in the first stage (Column L) and cutting patterns created for these leftovers (Column J'). The last three columns show the average computational time spent in each step (Columns "Patterns" and "Scenarios") and the total time (Column "Total"). For both steps, the maximum execution time was 1800 s. The complete description of all instances, including the cutting patterns created and scenarios evolved, is available at https://data.mendeley.com/datasets/tdr9jhyk62/1.

Table 3
Average number of created cutting patterns and computational time spent in the first two steps of the matheuristic.

Class	J	L	J'	Time (s)		
				Patterns	Scenarios	Total
3/Z/2/1	35.00	9.00	45.00	6.49	18.86	25.35
3/Z/2/2	42.00	9.40	47.00	8.78	20.23	29.01
3/M/2/1	31.30	6.90	34.50	3.92	17.49	21.41
3/M/2/2	35.50	7.30	36.50	3.79	19.67	23.46
5/Z/2/1	155.00	19.20	96.00	26.40	70.47	96.87
5/Z/2/2	186.00	17.40	87.00	26.05	88.10	114.15
5/M/2/1	134.80	12.60	63.00	12.15	56.87	69.02
5/M/2/2	160.40	14.30	71.50	14.50	72.02	86.52
8/Z/2/1	1274.20	29.70	148.50	297.77	776.85	1074.62
8/Z/2/2	1526.80	27.10	135.50	286.56	1077.96	1364.51
8/M/2/1	925.30	20.90	104.50	86.97	471.12	558.09
8/M/2/2	1089.50	20.50	102.50	106.62	613.78	720.40
10/Z/2/1	5044.40	31.20	156.00	1198.61	1800.00	2998.61
10/Z/2/2	6050.20	29.60	148.00	1479.38	1800.00	3279.38
10/M/2/1	3468.60	25.20	126.00	711.38	1800.00	2511.38
10/M/2/2	3330.90	26.00	130.00	937.89	1800.00	2737.89

Table 3 shows a significant increase in the number of cutting patterns for classes with 8 and 10 types of items. The variation of this number directly impacted the computational time of the scenario evolution, mainly with $I = 10$, since the co-evolutionary BRKGA algorithm reached the maximum execution time for all instances of these classes.

However, the higher number of cutting patterns allowed the third step of the matheuristic to find solutions with lower percentages of waste, as seen in Table 4, which presents the results obtained in the resolution of the stochastic model directly by CPLEX and by the Rounded L-shaped heuristic. In both cases, the maximum number of generated leftovers and their reuse rate were set to $U = 10$ and $\alpha = 80\%$, and the maximum execution time was 7200 s. Regarding the Rounded L-shaped heuristic, this time was divided between the two steps of the procedure, as both had a high computational time for classes with 8 and 10 types of items. The first step, referring to the resolution of M^{st}_{relax} by the L-shaped method, was interrupted after 3600 s of execution. Considering T_{relax} the time spent in this step, and T^{max}_{fix} the maximum time allowed for solving M^{st}_{fix} for each scenario, T^{max}_{fix} is calculated as follows:

$$T^{max}_{fix} = (7200 - T_{relax})/\Xi \tag{28}$$

The objective of the calculation described in (28) was to divide the remaining execution time equally among all Ξ scenarios. Furthermore, the resolution of M^{st}_{fix} was interrupted whenever the solution obtained by CPLEX had a gap less than or equal to 1%. We established this tolerance to reduce the time spent solving sub-problems that already had high-quality solutions, even without the certainty of their optimality.

In Table 4, the average values of the objective function (Column "OF"), the percentage of waste (Column "Waste (%)"), and the computational time in seconds (Column "Time (s)") are presented. Regarding the resolution of the stochastic model by CPLEX, the average gap provided by the solver (Column "Gap (%)") is also shown. If the average gap for a given class is greater than 0, it indicates that at least one of the instances of that class has reached the maximum execution time. The full detailed results for each instance of all classes are available in the online Supplementary Material provided with this paper.

Table 4 shows that CPLEX obtained the best solutions for all classes with $I = 3$. However, compared with the solutions from the Rounded L-shaped heuristic, the difference in the percentage of loss was, at most, 0.18%. Furthermore, the heuristic average computational time was approximately 7 s, while the resolution by CPLEX spent, on average, 5750 s.

Table 4
Average results of solving the stochastic model by CPLEX and by the Rounded L-shaped heuristic.

Class	CPLEX				*Rounded L-shaped*		
	OF	Waste (%)	Time (s)	Gap (%)	OF	Waste (%)	Time (s)
3/Z/2/1	**3 807.95**	**13.08**	5783.89	4.30	3 882.95	13.26	6.98
3/Z/2/2	**3 415.54**	**12.47**	7200.00	2.84	3 421.48	12.48	7.05
3/M/2/1	**19 056.20**	**15.31**	4264.05	0.29	19 071.77	15.32	6.85
3/M/2/2	**19 001.85**	**15.10**	5762.78	0.56	19 065.67	15.13	6.96
5/Z/2/1	5 395.71	10.75	7200.00	33.51	**4 719.76**	**9.62**	16.23
5/Z/2/2	4 613.21	11.78	7200.00	39.32	**4 132.43**	**10.55**	23.72
5/M/2/1	22 915.69	13.63	7200.00	2.75	**22 831.45**	**13.55**	38.42
5/M/2/2	15 628.37	10.31	7200.00	4.78	**15 549.28**	**10.23**	33.03
8/Z/2/1	5 288.22	8.91	7200.00	65.15	**3 060.27**	**5.37**	2916.54
8/Z/2/2	4 799.21	8.45	7200.00	65.56	**2 842.83**	**5.14**	3113.78
8/M/2/1	22 304.03	9.74	7200.00	9.65	**21 030.30**	**9.23**	1857.91
8/M/2/2	18 807.09	9.22	7200.00	10.58	**17 840.11**	**8.76**	1407.62
10/Z/2/1	7 966.49	8.84	7200.00	79.08	**3 687.00**	**4.32**	6766.79
10/Z/2/2	7 700.75	9.48	7200.00	74.18	**3 938.82**	**4.92**	6730.55
10/M/2/1	18 608.29	7.12	7200.00	26.89	**15 482.61**	**5.90**	5460.93
10/M/2/2	18 640.73	6.70	7200.00	24.35	**15 521.53**	**5.51**	4780.58

Table 5
Results for a single instance from each class with different time limits.

Class	CPLEX - 7200 (s)			CPLEX - 43200 (s)			*Rounded L-shaped*		
	OF	Waste (%)	Gap (%)	OF	Waste (%)	Gap (%)	OF	Waste (%)	Time (s)
3/Z/2/1	6 213.03	14.72	9.62	6 213.03	14.72	7.95	**6 201.00**	**14.67**	8.75
3/Z/2/2	**3 048.57**	**11.92**	5.10	**3 048.57**	**11.92**	4.03	3 050.02	12.07	6.92
3/M/2/1	**10 715.58**	**11.57**	1.17	**10 715.58**	**11.57**	0.86	**10 715.58**	**11.57**	7.01
3/M/2/2	**9 961.17**	**12.37**	1.33	**9 961.17**	**12.37**	1.15	10 071.08	12.41	8.06
5/Z/2/1	4 286.44	9.63	60.21	4 285.77	9.63	56.47	**4 076.90**	**9.39**	17.19
5/Z/2/2	2 420.66	6.14	67.17	1 919.10	5.01	56.75	**1 814.82**	**4.76**	59.95
5/M/2/1	20 974.92	12.16	4.78	20 877.24	12.11	4.27	**20 403.05**	**11.87**	15.52
5/M/2/2	7 095.09	5.83	14.96	**6 354.84**	**5.26**	4.76	6 617.53	5.46	17.55
8/Z/2/1	4 428.04	9.05	84.74	4 017.07	8.31	82.02	**1 949.42**	**4.40**	3517.67
8/Z/2/2	4 363.52	10.48	84.54	2 079.276	5.27	66.06	**1 462.13**	**3.85**	4380.08
8/M/2/1	20 120.43	7.29	17.03	19 918.48	7.23	15.88	**17 859.06**	**6.51**	2739.36
8/M/2/2	11 689.49	5.45	24.75	10 702.47	5.01	17.67	**9 828.21**	**4.62**	3740.25
10/Z/2/1	5 811.22	8.09	90.17	5 108.72	7.21	88.46	**2 527.61**	**3.69**	7137.28
10/Z/2/2	10 070.05	14.26	87.68	7 878.20	11.50	83.38	**3 487.99**	**5.54**	6153.95
10/M/2/1	9 252.87	6.67	47.92	8 100.24	5.86	40.36	**6 187.91**	**4.53**	7085.63
10/M/2/2	11 767.02	5.75	41.67	11 535.88	5.64	39.88	**8 634.08**	**4.28**	6712.19

For all classes with 5, 8, and 10 item types, CPLEX presented worse solutions than the Rounded L-shaped and reached the maximum execution time for all instances. The highest gap values, between 33% and 79%, were for classes with small items since they had a higher number of cutting patterns. The best solutions obtained by the heuristic refer to classes 10/Z/2/1 and 10/Z/2/2, for which the waste was reduced by half. However, the computational time was high, exceeding 6700 s.

Aiming to verify the behavior of our exact approach with a higher time limit, we performed computational experiments with 16 instances, one from each class, increasing the time limit to 43 200 s. We selected the instances whose solutions found by CPLEX had the largest gaps after 7200 s, i.e., the instances for which CPLEX could potentially benefit more from this time extension. The results presented in Table 5 show that only one solution was significantly improved (class 5/M/2/2). Although not comprehensive, the experiments indicate that this strategy would not be significantly advantageous.

We also performed additional experiments by solving two instances from Classes 3/Z/2/1 and 5/Z/2/1 with $S = 20$ types of standard plates in stock. From a practical point of view, considering a large number of plate types is not usual or even frequent. In the context studied, companies often prefer to handle a smaller number of plate types due to stock and storage efficiency and cost. Nevertheless, from an academic point of view, these experiments allow us to verify if it is feasible to use the proposed Rounded L-shaped heuristic to solve larger problems.

Tables 6 and 7 present the results of the additional experiments. These tables have the same columns as Tables 3 and 4. Since the number of possible leftovers for the instances with $S = 20$ was very high, we set the parameter N equal to 2. The results in Table 7 show that, by considering 20 types of standard plates, it was possible to considerably reduce the percentage of waste for both instances, mainly in the solutions obtained by the Rounded L-shaped heuristic. It was not possible to perform these additional experiments for instances with 8 and 10 types of items due to the extremely large number of cutting patterns created.

Table 6
Number of cutting patterns and computational time spent in the first two steps of the matheuristic for instances with $S = 2$ and 20.

Class	S	J	L	J'	Time (s)		
					Patterns	Scenarios	Total
3/Z/2/1	2	35	8	40	6.30	17.93	24.23
	20	287	82	164	30.26	162.73	192.99
5/Z/2/1	2	155	17	85	16.64	48.34	64.98
	20	1271	130	260	113.81	717.06	830.87

5.1. Measure of the uncertainty effects

Two metrics were used to assess the impact of uncertainty on the quality of the solutions obtained for the 2D-CSPULUD: the *Expected Value of Perfect Information (EVPI)* and the *Value of the Stochastic Solution (VSS)* (Birge, 1982). The EVPI measures the maximum amount the production manager is willing to pay in exchange for complete and accurate information about the future, the so-called *wait-and-see* (WS) values. Although perfect information cannot realistically be achieved,

Table 7
Results for instances with $S = 2$ and 20.

		CPLEX				*Rounded L-shaped*		
Class	S	OF	Waste (%)	Time (s)	Gap (%)	OF	Waste (%)	Time (s)
3/Z/2/1	2	6213.03	14.72	7200.00	9.62	6201.00	14.67	8.75
	20	3173.03	7.43	7200.00	50.99	1951.84	4.71	22.57
5/Z/2/1	2	5600.74	5.95	7200.00	11.66	5600.74	5.95	13.91
	20	4241.52	4.63	7200.00	52.06	2560.62	2.84	176.68

Table 8
Average values of the uncertainty impact measurement parameters.

Class	SP	WS	EVPI = SP-WS	$EVPI_{\%}$ = EVPI/WS	EEV	VSS = EEV-SP	$VSS_{\%}$ = VSS/SP
3/Z/2/1	3 806.75	3 486.61	320.13	**9.64**	4 060.56	253.81	**9.83**
3/Z/2/2	3 414.31	3 144.27	270.04	**9.15**	3 844.80	430.49	**11.39**
3/M/2/1	19 056.20	18 765.99	290.21	**1.78**	19 406.93	350.73	**1.91**
3/M/2/2	19 001.85	18 669.66	332.19	**2.19**	19 508.20	506.35	**2.84**
5/Z/2/1	4 719.76	4 189.23	530.54	**14.09**	5 106.58	386.82	**8.76**
5/Z/2/2	4 087.68	3 599.78	487.89	**17.41**	4 360.68	273.00	**10.88**
5/M/2/1	22 821.72	22 280.65	541.08	**3.00**	23 313.80	492.08	**3.60**
5/M/2/2	15 539.73	15 166.10	373.63	**3.68**	15 937.22	397.49	**2.99**

the EVPI measures the impact of the scenarios on the final value obtained for the problem. By considering known demand values, the problem can be solved avoiding costs with excess or lack of production. When addressing a minimization problem, $EVPI = SP - WS$, where SP represents the value of the solution obtained for the stochastic problem. As two methods were used to solve the proposed stochastic, SP received the lowest value among the solutions found by CPLEX and by the Rounded L-shaped heuristic. WS is the average value of the solutions for all scenarios when they are solved separately, knowing the demands in advance. Considering M_{ξ}^{WS} the solution value obtained by the stochastic model for scenario ξ only, we calculate WS as follows:

$$WS = \sum_{\xi=1}^{\Xi} \beta_{\xi} M_{\xi}^{WS}$$

The results presented by Nascimento, Cherri et al. (2022) are not directly comparable as they refer to a deterministic problem. However, since we used the instances from Nascimento, Cherri et al. (2022) by considering their original demands as the demands for the first-stage problem in the stochastic approach, we performed an analysis of the difference between the deterministic and stochastic problems by calculating the VSS, which measures the cost of not considering uncertainty in decision making, i.e., the waste resulting from replacing uncertain parameters with deterministic values. When addressing a minimization problem, $VSS = EEV - SP$, where EEV (*Expectation of using Expected Value*) is the expected result between scenarios of the application of a deterministic solution obtained from the resolution of a problem with an average scenario in the second stage.

Assuming d_i^{med} is the average demand of all scenarios, calculated as $d_i^{med} = \sum_{\xi=1}^{\Xi} \beta_{\xi} d'_{i\xi}$, a simplified problem is solved, by considering d_i^{med} to be the expected demand in the second stage. The solution value of this simplified problem is called EV (*Expected Value*). To calculate EEV, a deterministic problem M_{ξ}^{EEV} is solved for each scenario, fixing the values of first-stage variables as the values obtained by the simplified problem. EEV is calculated as follows, where $f^{M_{\xi}^{EEV}}$ is the solution value of each deterministic problem M_{ξ}^{EEV}:

$$EEV = EV + \sum_{\xi=1}^{\Xi} \beta_{\xi} f^{M_{\xi}^{EEV}}$$

The resolution of the sub-problems M_{ξ}^{WS} proved impracticable for classes with 8 and 10 types of items due to the high computational time. Therefore, we only analyzed the uncertainty effects when $I = 3$ or 5. Table 8 presents the average values of the parameters WS, $EVPI$, EEV and VSS for each class of instances.

The $EVPI$ values were higher for classes with small items, i.e., if we removed the uncertainty, these classes would have the most significant improvements in the quality of the solutions (between 9% and 17%). This information is useful to help managers to decide how much they will invest in better forecasting methods. Regarding VSS, the most significant values were also related to classes with small items, for which, considering uncertainty, the total waste was reduced by approximately 8%–11%. These values are significant and show the potential of using these methodologies in practice.

6. Conclusions

This research tackles the two-dimensional cutting stock problem with usable leftovers and uncertainty in demand by using a mathematical formulation that divides the problem into two stages. In the first stage, a cutting plan is defined to meet a known demand while the second stage adjusts the first stage decisions to optimize the use of the generated usable leftovers in a finite set of possible scenarios for the uncertain demand. Different strategies and solution methods were studied to solve this formulation, resulting in the development of a matheuristic that consists of creating a set of cutting patterns, generating scenarios for the demand and solving a stochastic model that defines the frequencies of the cutting patterns in both stages. By having this structure, the proposed matheuristic can be easily adapted to consider uncertainty in other parameters.

The two-stage stochastic model was solved through the CPLEX solver and a new heuristic procedure called Rounded L-shaped. This heuristic solves the first stage by the L-shaped method and it solves the second stage separately for each scenario with the first-stage decision variables fixed as the values obtained by the L-shaped. The results of the computational experiments with random instances from the literature showed that the Rounded L-shaped is more efficient in solving problems with more items. CPLEX only obtained the best solutions for the instances with three types of items but the average computational time was considerably long. In general, the performance of the proposed matheuristic was very satisfactory since it found good-quality solutions in an acceptable solving time.

For future research, adjustments can be made in the Rounded L-shaped heuristic to decrease the computational solving time for the instances with 8 and 10 types of items. Also, would be important to solve those instances (and the instances from classes 5/Z/2/1 and 5/Z/2/2) using other heuristic procedures to compare the results with the solutions obtained by the Rounded L-shaped heuristic, since the solutions provided by CPLEX have large gaps.

The two-stage formulation can be used to represent other stochastic problems, such as the multi-stage problem or the two-dimensional cutting stock problem with uncertainty in the dimension of usable leftovers due to the possibility of defects. In this case, some cutting patterns may be partially used or not used at all depending on the damaged area. To solve this problem, the proposed matheuristic can be adapted with changes only in the second and third steps since the strategy for building cutting patterns is independent of the stochastic parameter.

CRediT authorship contribution statement

Douglas Nogueira Nascimento: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Adriana Cristina Cherri:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Supervision. **José Fernando Oliveira:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Supervision. **Beatriz Brito Oliveira:** Conceptualization, Methodology, Validation, Formal analysis, Writing – original draft, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The full description of all instances used in the computational tests can be found at https://data.mendeley.com/datasets/tdr9jhyk62/1.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.cie.2023.109705.

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